

PROJECT PRESENTATION

FROM REGRESSION MODELS TO DEEP LEARNING: A JOURNEY THROUGH WIND SPEED PREDICTION TECHNIQUES"

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CONTENTS



- **ABSTRACT**
- **OBJECTIVE OF THE PROJECT & DATASET OVERVIEW**
- **EDA (EXPLORATORY DATA ANALYSIS)**
- **MODELS USED**
- **CALCULATION AND CASE STUDY**
- **LSTM (BEST MODEL)**
- **RESULTS**
- **CONCLUSION**

ABSTRACT

- In recent years, wind energy has emerged as a critical sustainable energy resource due to its eco-friendly, carbon-free, and non-polluting nature. Accurate wind speed forecasting plays a vital role in optimizing the use of wind energy for power generation.
- In this project, multiple machine learning models — including Linear Regression, K-Nearest Neighbors (KNN), Random Forest, and Long Short-Term Memory (LSTM) networks were implemented to predict wind speed based on meteorological features.
- Special attention was given to data preprocessing, where missing values were handled using different strategies: **dropping missing entries, forward fill, and mean fill methods.**
- The models were evaluated using statistical error metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).
- Among all the models, the LSTM network demonstrated the best performance, significantly outperforming traditional machine learning approaches. It captured the sequential patterns in the wind speed time series data effectively, providing highly accurate predictions.
- The results validated that deep learning approaches, specifically LSTM models combined with careful data preprocessing, offer a promising solution for reliable short-term wind speed forecasting.

OBJECTIVE OF THE PROJECT & DATASET OVERVIEW



 **Objective:** This project focuses on predicting wind speed using various machine learning models. The goal is to evaluate and compare their performances based on error metrics.

 Predict **Wind Speed (WIND)** using other meteorological features

- **Source:** Collected from Kaggle
<https://www.kaggle.com/code/majidmh/wind-speed-prediction-regression/input>
- **Dataset Size:** Contains 6574 entries (rows) & 9 columns
- **Target Variable:** WIND (Wind Speed)
- **Data Type:** Time-series tabular data with some missing values

Handling Missing Values

We used three strategies to handle missing data:

-  **Drop Missing Values** [6574-5638 = 936]
-  **Forward Fill**
-  **Mean Fill**

 Feature	 Description
 DATE	Date of the observation (not used in modeling)
 WIND	Target – Wind speed (continuous)
 RAIN	Rainfall amount
 T.MAX	Maximum daily temperature
 T.MIN	Minimum daily temperature
 T.MIN.G	Ground-level minimum temperature
 IND	Weather indicator (e.g., humidity/cloud cover)
 IND.1	Weather indicator (requires interpretation)
 IND.2	Weather indicator (requires interpretation)

EDA (EXPLORATORY DATA ANALYSIS)

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6574 entries, 0 to 6573
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   DATE         6574 non-null    object
1   WIND         6574 non-null    float64
2   IND         6574 non-null    int64
3   RAIN         6574 non-null    float64
4   IND.1        6513 non-null    float64
5   T.MAX        5953 non-null    float64
6   IND.2        6513 non-null    float64
7   T.MIN        5900 non-null    float64
8   T.MIN.G      6214 non-null    float64
dtypes: float64(7), int64(1), object(1)
memory usage: 462.4+ KB
None
```

[Method 1] Summary After Removing Missing Values:

	WIND	RAIN	T.MAX	T.MIN	T.MIN.G \
count	5638.000000	5638.000000	5638.000000	5638.000000	5638.000000
mean	9.682297	1.875647	13.285669	6.446949	2.757627
std	4.938009	3.973763	4.890483	4.626693	5.576713
min	0.000000	0.000000	-0.100000	-11.500000	-13.500000
25%	5.910000	0.000000	9.600000	3.100000	-1.000000
50%	9.080000	0.200000	13.100000	6.500000	3.000000
75%	12.830000	2.000000	17.200000	10.000000	7.000000
max	30.370000	67.000000	26.800000	18.000000	15.800000

	IND	IND.1	IND.2
count	5638.000000	5638.000000	5638.000000
mean	0.399610	0.012593	0.093828
std	1.189562	0.157654	0.303538
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	4.000000	2.000000	3.000000

[Method 2] Summary After Forward Fill (ffill):

	WIND	RAIN	T.MAX	T.MIN	T.MIN.G \
count	6574.000000	6574.000000	6574.000000	6574.000000	6574.000000
mean	9.796834	1.885169	13.407012	6.497216	2.751977
std	4.977272	4.030529	4.972564	4.782379	5.621692
min	0.000000	0.000000	-0.100000	-11.500000	-14.400000
25%	6.000000	0.000000	9.600000	2.900000	-1.000000
50%	9.210000	0.200000	13.500000	6.600000	3.000000
75%	12.960000	2.000000	17.500000	10.400000	7.000000
max	30.370000	67.000000	26.800000	18.000000	15.800000

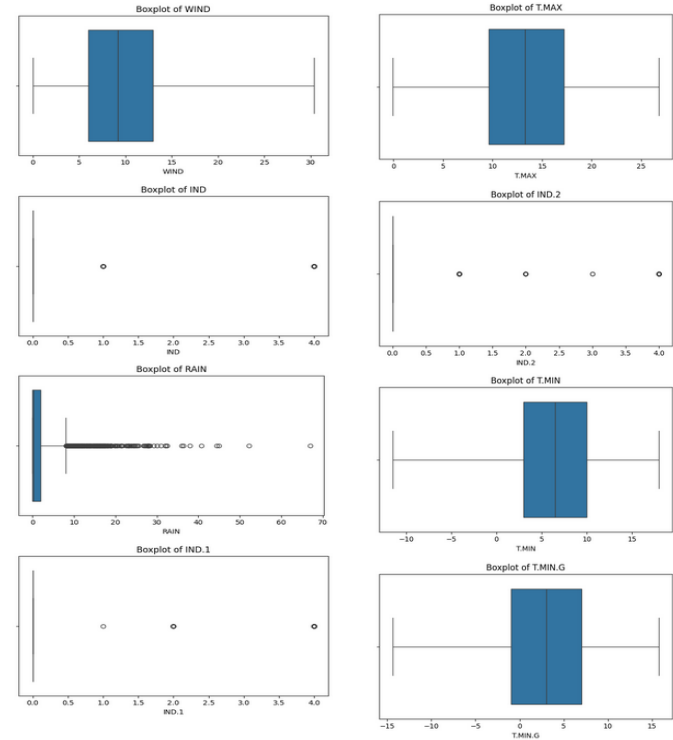
	IND	IND.1	IND.2
count	6574.000000	6574.000000	6574.000000
mean	0.391542	0.353057	0.460146
std	1.179092	1.123823	1.172904
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	4.000000	4.000000	4.000000

[Method 3 - Mean Fill Only] Summary After Filling Missing Values with Mean:

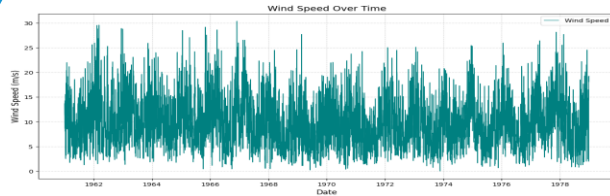
	WIND	RAIN	T.MAX	T.MIN	T.MIN.G \
count	6574.000000	6574.000000	6574.000000	6574.000000	6574.000000
mean	9.796834	1.885169	13.339123	6.411678	2.736547
std	4.977272	4.030529	4.653792	4.393061	5.414517
min	0.000000	0.000000	-0.100000	-11.500000	-14.400000
25%	6.000000	0.000000	10.000000	3.500000	-0.900000
50%	9.210000	0.200000	13.339123	6.411678	2.736547
75%	12.960000	2.000000	16.800000	9.600000	6.600000
max	30.370000	67.000000	26.800000	18.000000	15.800000

	IND	IND.1	IND.2
count	6574.000000	6574.000000	6574.000000
mean	0.391542	0.356364	0.464456
std	1.179092	1.123303	1.172094
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	4.000000	4.000000	4.000000

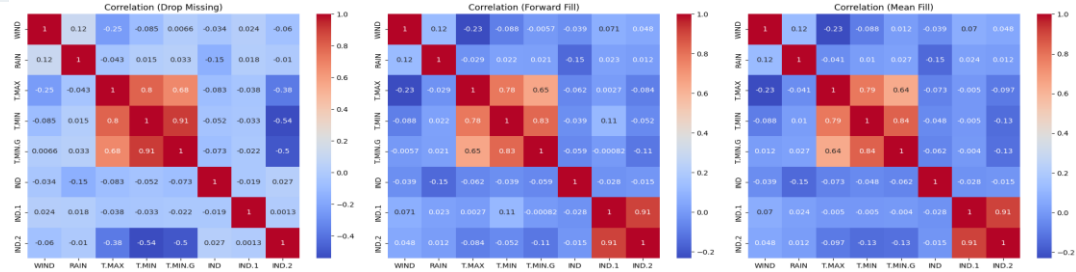
Outliers Check



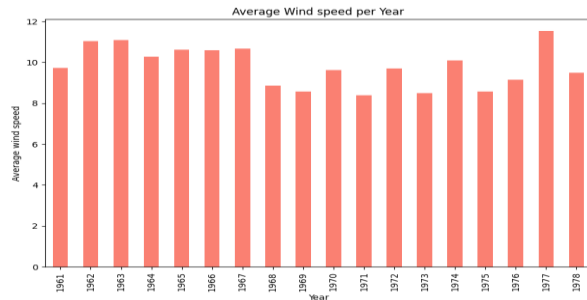
EDA (EXPLORATORY DATA ANALYSIS)



Handled missing values
Normalized input features
Feature selection
Train-test split for model evaluation

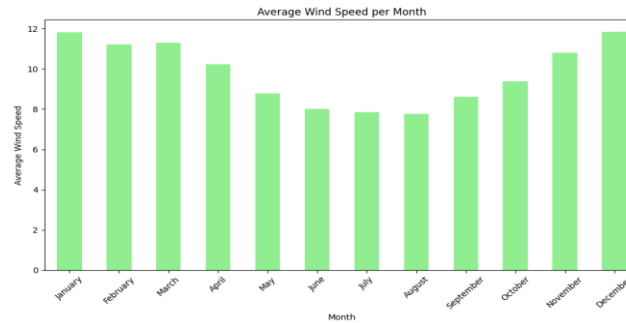


Overall, the **correlation analysis** shows that temperature features, especially **T.MAX**, have a mild **positive relationship** with **WIND**, justifying their use in model building, while different missing data handling methods slightly impact the strength of these relationships



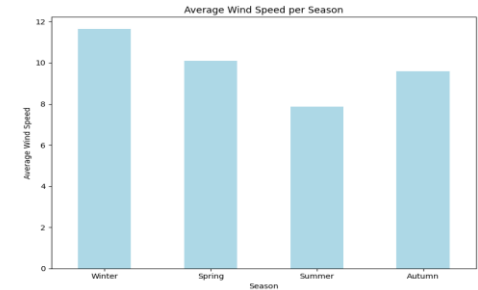
Yearly Trend

Average wind speed remains relatively stable from 1961 to 1978.
From the yearly analysis, **1977 recorded the highest average** wind speed among all the years studied



Monthly Trend

Higher wind speeds during winter months (Dec, Jan, Feb).
Lower wind speeds during summer months (May, June, July).



Seasonal Trend

Winter has the **highest average** wind speed.
Summer has the **lowest average** wind speed.

MODELS USED



Linear Regression (1-Variable)

- **Feature Used:** T.MAX (Maximum Temperature)
- **Goal:** Understand how changes in maximum temperature alone influence wind speed. A basic regression model that uses only one feature (**T.MAX**).



Linear Regression (2-Variable)

- **Features Used:** T.MAX (Maximum Temperature) and RAIN (Rainfall)
- **Goal:** Analyze the combined linear influence of **temperature and rainfall** on wind speed. A multiple linear regression model with **two independent variables**.



Linear Regression (Multiple Variables)

- **Features Used:** All input features (**excluding WIND and DATE**)
- **Goal:** Capture the combined linear effect of **multiple weather variables** on wind speed. A **multivariate linear regression model** that considers several predictors simultaneously.



K-Nearest Neighbors (KNN)

- **Features Used:** All input features (**excluding WIND and DATE**)
- **Goal:** Predict wind speed by averaging the wind speeds of the **k most similar historical weather entries**.



Random Forest Regressor

- **Features Used:** All available features (**excluding WIND and DATE**).
- **Goal:** Predict wind speed based on weather variables. Models the **non-linear** relationships between **weather features and wind speed**.



LSTM (Long Short-Term Memory) Network

- **Features Used:** All input features (**excluding WIND and DATE**)
- **Goal:** Forecast wind speed by learning temporal patterns in historical weather data. An advanced type of **Recurrent Neural Network (RNN)** designed to handle sequential data and long-term dependencies

Note: All models will apply on three preprocessed datasets (**Drop Missing, Forward Fill, Mean Fill**) to compare their performance consistently.
Data Split: 80% training, 20% testing split, ensuring reliable evaluation of the model's performance.

- **Multiple models** were implemented to predict wind speed, including Linear Regression (1-variable/2-variable/Multi-variables), Random Forest, KNN, and LSTM (Long Short-Term Memory).
- These models were evaluated across three different versions of the dataset, preprocessed using **Drop Missing, Forward Fill, and Mean Fill** methods.
- Each model was assessed using metrics such as **MSE, RMSE, and MAE** to compare their performance objectively.

LINEAR REGRESSION (1-VARIABLE/2-VARIABLES/MULTI-VARIABLES) OBSERVATIONS



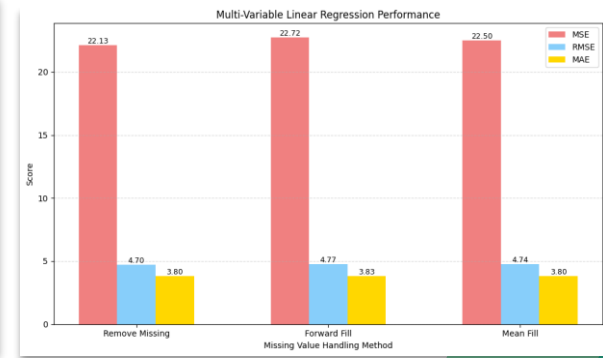
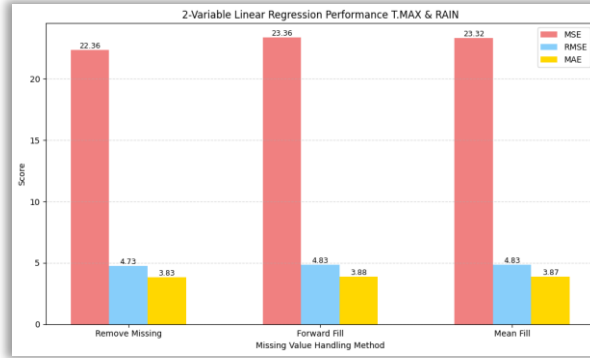
Model Type	Method Used	MSE	RMSE	MAE
1-Variable	Remove Missing	22.8577	4.7810	3.8775
	Forward Fill	23.7905	4.8776	3.9192
	Mean Fill	23.7367	4.8720	3.9026
2-Variable	Remove Missing	22.3561	4.7282	3.8331
	Forward Fill	23.3594	4.8332	3.8802
	Mean Fill	23.3231	4.8294	3.8653
Multi-Variables	Remove Missing	22.1276	4.7040	3.8037
	Forward Fill	22.7181	4.7663	3.8310
	Mean Fill	22.4957	4.7430	3.7985

- **Three Models Evaluated:** 1-Variable (T.MAX), 2-Variable (T.MAX, RAIN) and Multi-Variable (T.MAX, RAIN, other features)
- **Three Data Handling Methods:** Remove Missing Values, Forward Fill, Mean Fill.
- **Evaluation Metrics Used:**
MSE (Mean Squared Error): Penalizes large errors.
RMSE (Root MSE): Interpretable in original units.
MAE (Mean Absolute Error): Measures average absolute error.
- **Best Model:** ☒ Multi-Variable Regression with 'Remove Missing' method
MSE: **22.13**, RMSE: **4.70** and mean fill - MAE: **3.79**
- **Impact of Feature Addition:** Errors decrease as more features are included → **better predictions.**
- **Imputation Comparison:** Removing missing values gives consistently better results than forward/mean fill.
- Remove Missing gives the best overall performance across all models.
- **Mean Fill** performs **better than** Forward Fill in most models:
Slightly **lower MSE, RMSE, and MAE values.**
- Mean Fill is especially useful when we can't afford to lose data by removing rows.

LINEAR REGRESSION (1-VARIABLE/2-VARIABLES/MULTI-VARIABLES) GRAPHICAL REPRESENTATION AND RESULTS



Conclusion: More features + proper data cleaning → better model accuracy.



```
1-Variable Linear Regression Performance (T.MAX)
Method: Remove Missing Values
MSE: 22.8577
MAE: 3.8775
RMSE: 4.7810
-----
Method: Forward Fill
MSE: 23.7905
MAE: 3.9192
RMSE: 4.8776
-----
Method: Mean Fill
MSE: 23.7367
MAE: 3.9026
RMSE: 4.8720
-----
```

```
2-Variable Linear Regression Performance (T.MAX) & (RAIN)
Method: Remove Missing
MSE: 22.3561
RMSE: 4.7282
MAE: 3.8331
-----
Method: Forward Fill
MSE: 23.3594
RMSE: 4.8332
MAE: 3.8802
-----
Method: Mean Fill
MSE: 23.3231
RMSE: 4.8294
MAE: 3.8653
-----
```

```
Multi-Variable Linear Regression Performance
Method: Remove Missing
MSE: 22.1276
RMSE: 4.7040
MAE: 3.8037
-----
Method: Forward Fill
MSE: 22.7181
RMSE: 4.7663
MAE: 3.8310
-----
Method: Mean Fill
MSE: 22.4957
RMSE: 4.7430
MAE: 3.7985
-----
```

KNN & RANDOM FOREST REGRESSION OBSERVATIONS



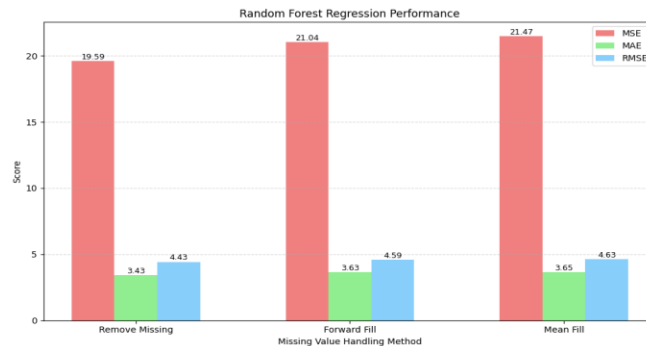
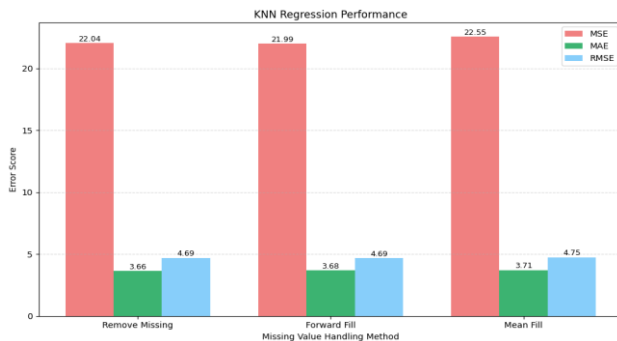
Model	Method	MSE	RMSE	MAE
KNN	Remove Missing	22.0425	4.6949	3.6591
	Forward Fill	21.9884	4.6892	3.6849
	Mean Fill	22.5459	4.7482	3.7094
Random Forest	Remove Missing	19.5884	4.4259	3.4315
	Forward Fill	21.0440	4.5874	3.6299
	Mean Fill	21.4673	4.6333	3.6484

- Random Forest **outperforms** KNN in all evaluation metrics.
- **Remove Missing method** consistently delivers the lowest error for both models.
- **Random Forest: Best MSE: 19.5884, Best RMSE: 4.4259, Best MAE: 3.4315**
- **KNN's performance** is less stable, slightly worse across all imputation methods.
- **Forward Fill > Mean Fill** for both models across all metrics.
- KNN has higher error and more sensitivity to missing data handling.
- **Random Forest with Remove Missing** is the best performing setup in this study.

KNN & RANDOM FOREST REGRESSION GRAPHICAL REPRESENTATION AND RESULTS



"**Random Forest** outperforms KNN overall; more features combined with proper data cleaning significantly **improve model accuracy**."



KNN Regression Performance:

Method: Remove Missing

MSE: 22.0425

MAE: 3.6591

RMSE: 4.6949

Method: Forward Fill

MSE: 21.9884

MAE: 3.6849

RMSE: 4.6892

Method: Mean Fill

MSE: 22.5459

MAE: 3.7894

RMSE: 4.7482

Random Forest Regression Performance:

Method: Remove Missing

MSE: 19.5884

MAE: 3.4315

RMSE: 4.4259

Method: Forward Fill

MSE: 21.0440

MAE: 3.6299

RMSE: 4.5874

Method: Mean Fill

MSE: 21.4673

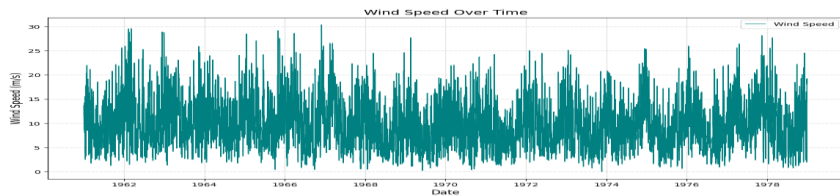
MAE: 3.6484

RMSE: 4.6333

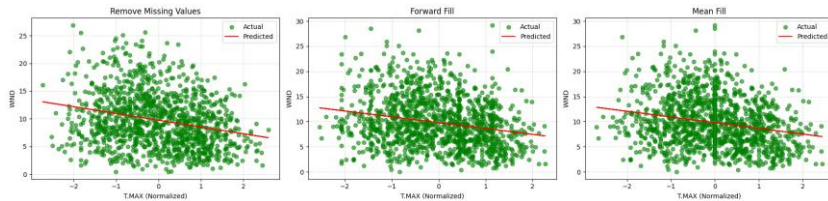
ACTUAL VS PREDICTED (TEST DATA PREDICTIONS) MODEL-TO-MODEL COMPARISON



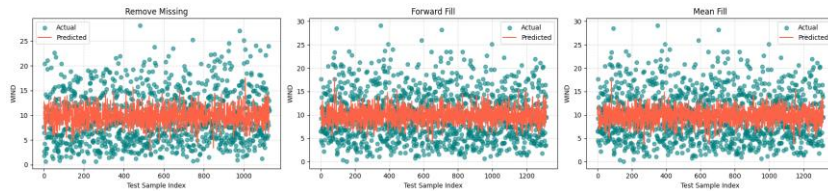
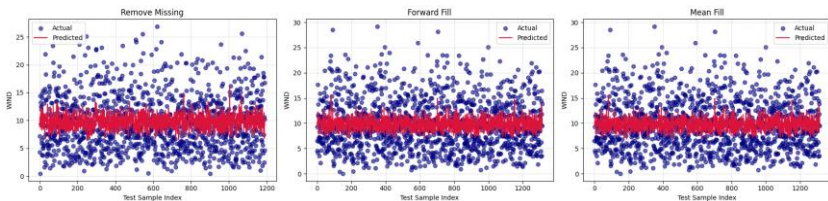
Regression Predictions on Test Data
Multi-variable (7 features → WIND)



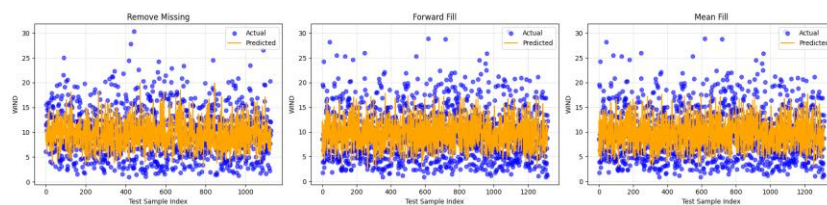
Linear Regression (1-variable) Fit on Test Data (TMAX → WIND)



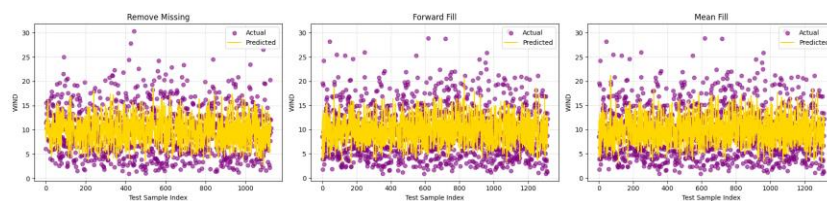
Linear Regression (2-Variable) Fit on Test Data (TMAX & RAIN → WIND)



KNN Regression Predictions vs Actuals



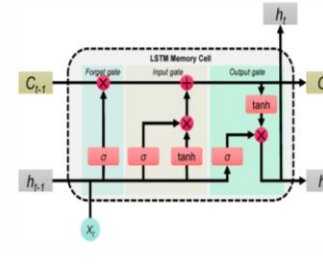
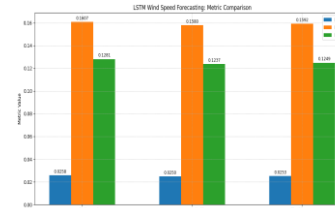
Random Forest Regression Predictions vs Actuals



LSTM(BEST MODEL)



- **LSTM (Long Short-Term Memory)** is a type of **Recurrent Neural Network (RNN)**, specifically designed to model sequential data or time-series data.
- LSTMs have **memory cells** that store information over time. These cells help the model **retain important data** and "forget" irrelevant information, allowing it to learn patterns that span across long sequences.
- LSTM outperforms traditional models like **Linear Regression, KNN and Random Forest Regression** in terms metrics across all data imputation strategies.
- **Improved Prediction Accuracy:** The LSTM model significantly reduces the error compared to linear models and KNN.
- **LSTM Performance:** The predicted values closely track the actual values.

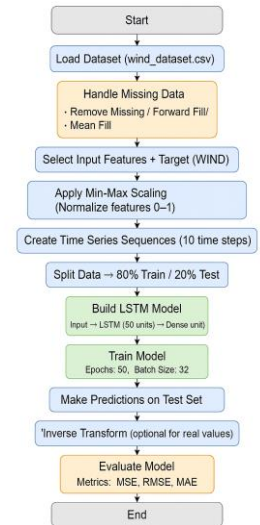


```

Final Evaluation Metrics:
Strategy: Remove Missing Values
RMSE: 0.1807
MAE: 0.1104
MAPE: 0.1001

Strategy: Forward Fill
RMSE: 0.1806
MAE: 0.1103
MAPE: 0.1000

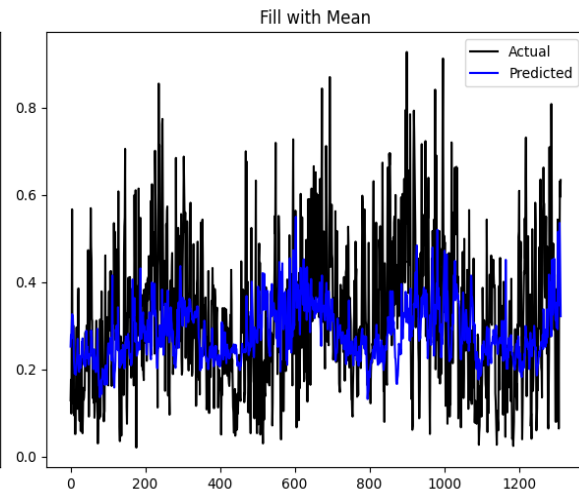
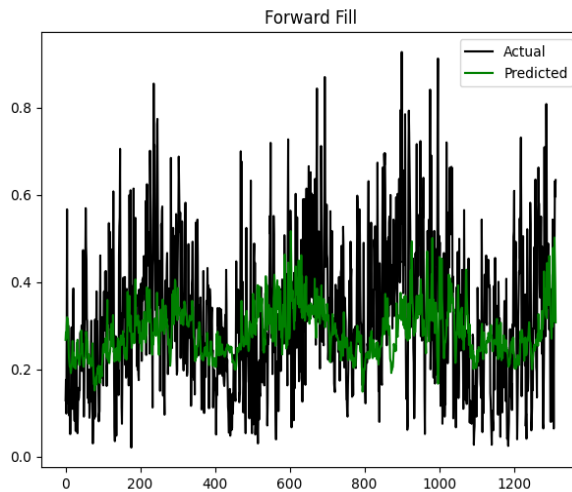
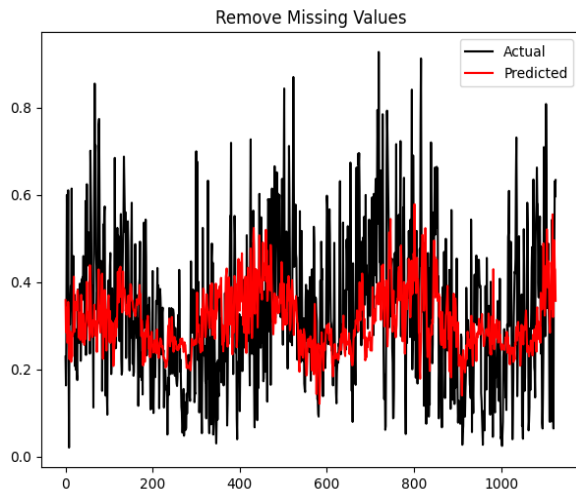
Strategy: Fill with Mean
RMSE: 0.1806
MAE: 0.1103
MAPE: 0.1000
  
```



LSTM MODEL EVALUATION: ACTUAL VS PREDICTED



Actual vs Predicted Wind Speed (Side-by-Side)

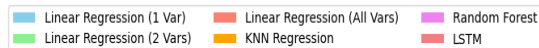


STATISTICAL ERROR VALUES OF ALL APPROACHES

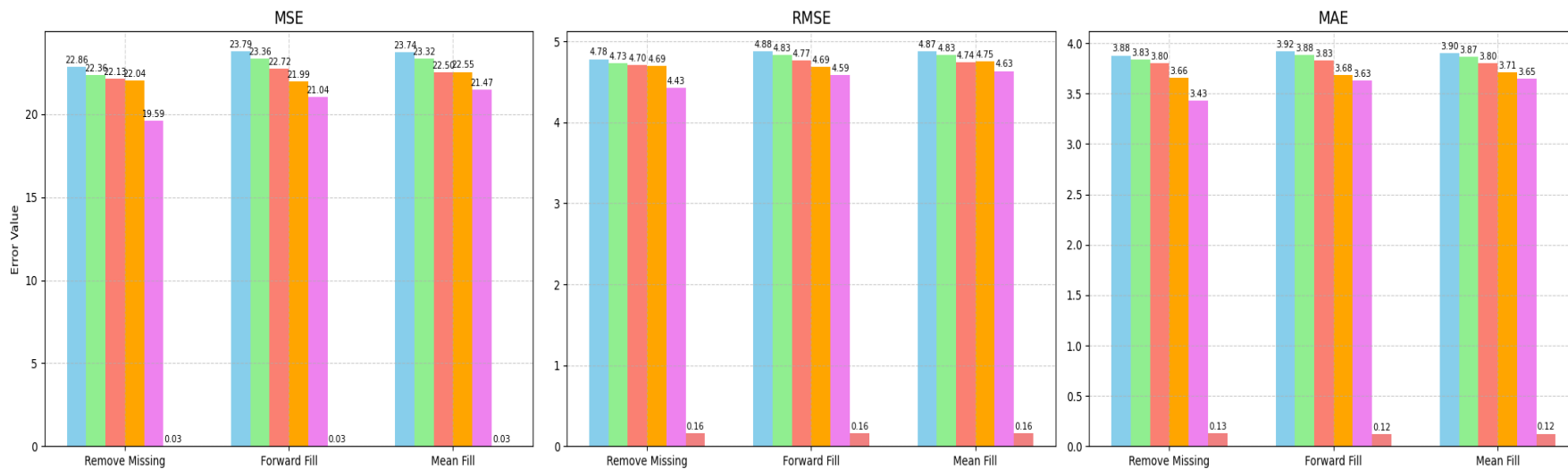


Model	MSE (Remove)	MSE (Forward Fill)	MSE (Mean Fill)	MAE (Remove)	MAE (Forward Fill)	MAE (Mean Fill)	RMSE (Remove)	RMSE (Forward Fill)	RMSE (Mean Fill)
Linear (1 Var)	22.8577	23.7905	23.7367	3.8775	3.9192	3.9026	4.7810	4.8776	4.8720
Linear (2 Var)	22.3561	23.3594	23.3231	3.8331	3.8802	3.8653	4.7282	4.8332	4.8294
Linear (Multi Var)	22.1276	22.7181	22.4957	3.8037	3.8310	3.7985	4.7040	4.7663	4.7430
KNN	22.0425	21.9884	22.5459	3.6591	3.6849	3.7094	4.6949	4.6892	4.7482
Random Forest	19.5884	21.0440	21.4673	3.4315	3.6299	3.6484	4.4259	4.5874	4.6333
LSTM	0.0258	0.0250	0.0253	0.1281	0.1237	0.1249	0.1607	0.1580	0.1592

GRAPHICAL REPRESENTATION OF ALL MODELS ON DIFFERENT DATASETS



Model Performance Comparison Across Missing Data Strategies



BEST MODEL (LSTM)



=== Performance Comparison Table ===

	MSE	RMSE	MAE
Linear Regression (1 Var) - Remove Missing	22.8577	4.7810	3.8775
Linear Regression (1 Var) - Forward Fill	23.7905	4.8776	3.9192
Linear Regression (1 Var) - Mean Fill	23.7367	4.8720	3.9026
Linear Regression (2 Vars) - Remove Missing	22.3561	4.7282	3.8331
Linear Regression (2 Vars) - Forward Fill	23.3594	4.8332	3.8802
Linear Regression (2 Vars) - Mean Fill	23.3231	4.8294	3.8653
Linear Regression (All Vars) - Remove Missing	22.1276	4.7040	3.8037
Linear Regression (All Vars) - Forward Fill	22.7181	4.7663	3.8310
Linear Regression (All Vars) - Mean Fill	22.4957	4.7430	3.7985
KNN Regression - Remove Missing	22.0425	4.6949	3.6591
KNN Regression - Forward Fill	21.9884	4.6892	3.6849
KNN Regression - Mean Fill	22.5459	4.7482	3.7094
Random Forest - Remove Missing	19.5884	4.4259	3.4315
Random Forest - Forward Fill	21.0440	4.5874	3.6299
Random Forest - Mean Fill	21.4673	4.6333	3.6484
LSTM - Remove Missing	0.0258	0.1607	0.1281
LSTM - Forward Fill	0.0250	0.1580	0.1237
LSTM - Mean Fill	0.0253	0.1592	0.1249

Handwritten note: "Best" with an arrow pointing to the LSTM - Forward Fill row.

=== Performance Comparison Table ===

Model - Method	MSE	RMSE	MAE
Linear Regression (1 Var) - Forward Fill	23.7905	4.8776	3.9192
Linear Regression (1 Var) - Mean Fill	23.7367	4.8720	3.9026
Linear Regression (2 Vars) - Forward Fill	23.3594	4.8332	3.8802
Linear Regression (2 Vars) - Mean Fill	23.3231	4.8294	3.8653
Linear Regression (1 Var) - Remove Missing	22.8577	4.7810	3.8775
Linear Regression (All Vars) - Forward Fill	22.7181	4.7663	3.8310
Linear Regression (All Vars) - Mean Fill	22.4957	4.7430	3.7985
KNN Regression - Mean Fill	22.5459	4.7482	3.7094
Linear Regression (2 Vars) - Remove Missing	22.3561	4.7282	3.8331
Linear Regression (All Vars) - Remove Missing	22.1276	4.7040	3.8037
KNN Regression - Remove Missing	22.0425	4.6949	3.6591
KNN Regression - Forward Fill	21.9884	4.6892	3.6849
Random Forest - Mean Fill	21.4673	4.6333	3.6484
Random Forest - Forward Fill	21.0440	4.5874	3.6299
Random Forest - Remove Missing	19.5884	4.4259	3.4315
LSTM - Remove Missing	0.0258	0.1607	0.1281
LSTM - Mean Fill	0.0253	0.1592	0.1249
LSTM - Forward Fill	0.0250	0.1580	0.1237

Best Model: LSTM (Long Short-Term Memory)

- The LSTM model **outperformed all other models** across all dataset versions.
- LSTM achieved the **lowest values of MSE, RMSE, and MAE** compared to all other models.
- Its performance was consistently **better across all three preprocessed datasets (Drop Missing, Forward Fill, Mean Fill)**.
- Especially when using **Forward Fill**, LSTM handled sequences smoothly without abrupt gaps.

RESULTS AND CONCLUSION



- Multiple models were tested for wind speed prediction
- **LSTM (Long Short-Term Memory)** emerged as the best model across all dataset versions.
- Achieved the **lowest error metrics (MSE, RMSE, MAE)** consistently across:
Drop Missing, Forward Fill, Mean Fill
- **Forward Fill preprocessing + LSTM** delivered the **best overall performance**, ensuring smooth sequence continuity.
- LSTM is the most reliable and precise model for wind speed forecasting in this project.

Recommendation: Use **LSTM combined with Forward Fill** for future wind speed prediction tasks.

FUTURE WORK

- **Apply EEMD-LSTM:** Combine signal decomposition with LSTM for better forecasting.
- Include **wind direction** and **more environmental** signals for better accuracy.
- Apply hybrid models (e.g., **EEMD + LSTM**, **CNN + LSTM**).
- Deploy as a web-based wind forecasting tool for energy.



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THANK YOU

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